$$\pi = \dfrac{\dfrac{\overset{\pi_1^L}{\vdash F\{y/x\}, \Gamma, I_1} \qquad \overset{\pi_1^R}{\vdash I_1^{\perp}, \Delta}}{\vdash F\{y/x\}, \Gamma, \Delta}\ (\exists)}{\vdash \exists x F, \Gamma, \Delta}\ (\mathsf{Cut})$$